Revision of Week 3

* Random Variable $X : \Omega \longrightarrow \mathbb{R}$ ((Ω, F, P))
* Discrete Random Variable S_X is finite or countable
* Cumulative Distribution Function (CDF) $F_X(x) = P[X \le x]$
* Probability mass function (PMF) $\boxed{P_X(k) = P(X = k)} \in [0,1]$
  - Bernoulli Random variable $\{0,1\}$ $P(X=1)=p$ $P(X=0)=1-p$
  - Binomial Random Variable $\{0,1,\dots,n\}$ $P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$
  - Geometric Random Variable $\{0,1,\dots\}$ $P_X(k) = (1-p)^{k-1} p$
  - Poisson Random Variable $\{0,1,2,\dots\}$ $P_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$
* Examples of Binomial R.V.
* Continuous Random Variable $\longrightarrow$ $P[X = x] = 0$, $P[c \le x \le d] = \int_c^d f_X(x)\,dx$
  - Pdf $f_X : \mathbb{R} \to [0, \infty)$
* Uniform Random variable $f_X(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$
* Exponential Random Variable
* Gaussian Random Variable

sample space

$X : \Omega \longrightarrow \mathbb{R}$

$X(\omega_1) = x_1$ $X(\omega_2) = x_2$ ...

$S_X = \{ X(\omega) \mid \omega \in \Omega \}$ (ω: sample)

3 coins are tossed

$\Omega = \{HHH, HHT, HTH, \dots, TTT\}$ (ω)

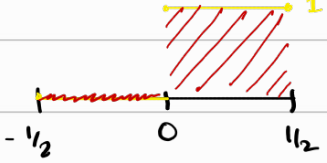
X = no. of Heads

Heads: 1, Tails: 0 $S_X = \{0,1\}$ $\{0,1,2,3\}$

Y : difference between no. of Heads and no. of tails

$X \sim U[0,1]$ $f_X(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{o.w} \end{cases}$

$P[\frac{1}{2} \le x \le \frac{3}{4}] = \int_{1/2}^{3/4} (1)\,dx = [x]_{1/2}^{3/4} = \frac{1}{4}$



$P[-\frac{1}{2} \le x \le \frac{1}{2}] = \int_{-1/2}^{1/2} f_X(x)\,dx = \int_{-1/2}^{0} 0\,dx + \int_{0}^{1/2} 1\,dx = 0 + (\frac{1}{2} - 0) = \frac{1}{2}$

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4. The distribution function of the random variable X is given

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} & 0 \le x < 1 \\ \frac{2}{3} & 1 \le x < 2 \\ \frac{11}{12} & 2 \le x < 3 \\ 1 & 3 \le x \end{cases}$$

(a) Plot this distribution function.

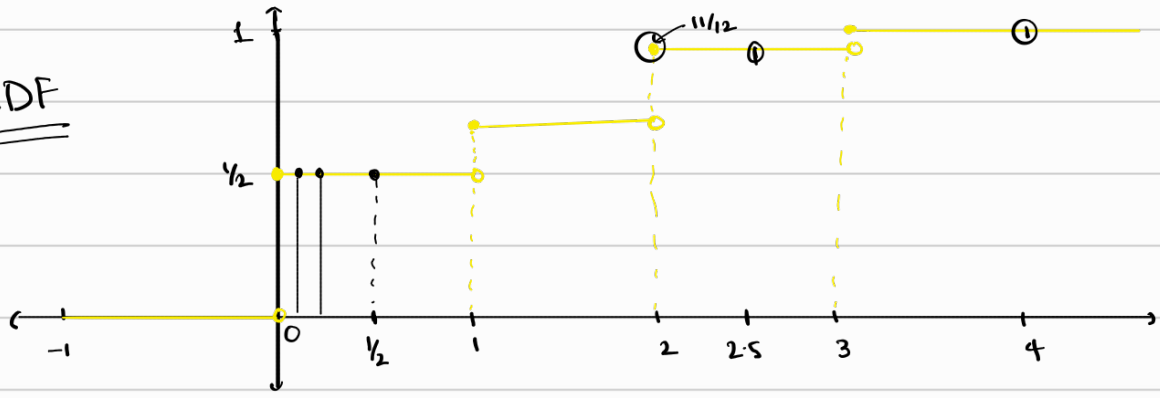
(b) What is $P\{X > \frac{1}{2}\}$? $= 1 - P(X \le 1/2) = 1 - 1/2 = 1/2$

(c) What is $P\{2 < X \le 4\}$? $F_X(4) - F_X(2) = 1 - 11/12 = 1/12$

(d) What is $P\{X < 3\}$? $1 - P[X \ge 3] = F_X(3) - P[X = 3] = 1 - \frac{1}{12} = \frac{11}{12}$

(e) What is $P\{X = 1\}$?

CDF



$1/2 = F_X(1/2) = P[X \le 1/2]$ $\qquad F_X(0.2) = 1/2$ $\qquad F_X(0.001) = 1/2$

PMF

$P(X=0) = F_X(0) - F_X(0^-) = 1/2 - 0 = 1/2$

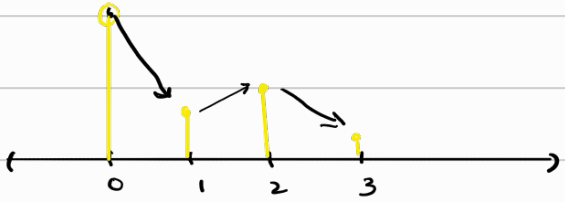
$P(X=1) = F_X(1) - F_X(1^-) = 2/3 - 1/2 = 1/6$

$P(X=2) = F_X(2) - F_X(2^-) = 11/12 - 2/3 = 1/4$

$P(X=3) = F_X(3) - F_X(3^-) = 1 - 11/12 = 1/12$

1

PMF



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1. Five men and 5 women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman (for instance, $X = 2$ if the top-ranked person was male and the next-ranked person was female). Find $P\{X = i\}, i = 1, 2, 3, \ldots, 8, 9, 10.$

$P(X=10) = 0 = P(X=9) = P(X=8) = P(X=7)$

$P(X=1) = \frac{\text{no. of women}}{\text{Total no. of people}} = \frac{5}{10} = 0.5$

F _____
↑
$x=1$

$P(X=2) = \frac{5}{10} \times \frac{5}{9}$

MF
↑
$x=2$

$P(X=3) = \frac{5}{10} \times \frac{4}{9} \times \frac{5}{8}$

MMF
↑
$x=3$

$P(X=4) = \frac{5}{10} \times \frac{4}{9} \times \frac{3}{8} \times \frac{5}{7}$

MMMF
↑
$x=4$

$P(X=5) = \frac{5}{10} \times \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{5}{6}$ → MMMMF M
↑
$x=5$

$P(X=6) = \frac{5 \times 4 \times 3 \times 2 \times 1}{10 \times 9 \times 8 \times 7 \times 6}$

MMMMM F F F F F
↑
$x=6$

$P(X=7) = \frac{5 \times 4 \times 3 \times 2 \times 1 \times 0}{10 \times 9 \times 8 \times 7 \times 6 \times 5}$

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1. A satellite system consists of 4 components and can function adequately if at least 2 of the 4 components are in working condition. If each component is, independently, in working condition with probability .6, what is the probability that the system functions adequately?

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7. If X and Y are binomial random variables with respective parameters (n, p) and $(n, 1-p)$, verify and explain the following identities:

(a) $P\{X \le i\} = P\{Y \ge n-i\}$;

(b) $P\{X = k\} = P\{Y = n-k\}$.

$X \sim \text{Bin}(n, p)$ $\qquad$ $Y \sim \text{Bin}(n, 1-p)$

(a) LHS $= P(X \le i) = \sum_{k=0}^{i} \binom{n}{k} p^k (1-p)^{n-k}$

RHS $= P(Y \ge n-i) = \sum_{k=n-i}^{n} \binom{n}{k} (1-p)^k (p)^{n-k}$

$= \sum_{j=0}^{i} \binom{n}{j} (1-p)^{n-j} p^j$

$= $ LHS

$j = n-k \iff k = n-j$

$j = i$ for $k = n-i$

$j = i-1$ for $k = n-i+1$

$\vdots$

$j = 0$ for $k = n$

$\binom{n}{k} = \binom{n}{n-k} = \binom{n}{j}$

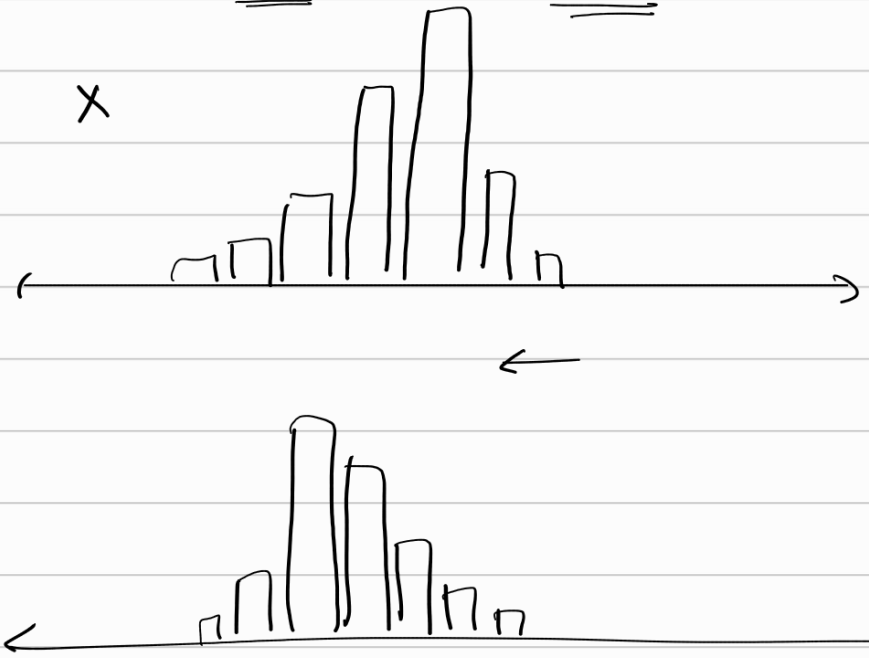
(b)

$P(X=k) = P(Y=n-k)$

$\binom{n}{k} p^k (1-p)^{n-k} = \binom{n}{n-k} (1-p)^{n-k} p^k$

$P(X=k) = P(Y=n-k)$

$X=k$ $\qquad$ $Y=n-k$



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4.13. A salesman has scheduled two appointments to sell encyclopedias. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs \$1000, or the standard model, which costs \$500. Determine the probability mass function of X, the total dollar value of all sales.

S_1 = Sale 1 $\qquad$ S_2 = Sale 2

A = deluxe model $\qquad$ B = Standard model

$$S_1 = \begin{cases} 0 & \text{if no sale} \\ 500 & \text{if sale B} \\ 1000 & \text{if sale A} \end{cases} \qquad S_2 = \begin{cases} 0 & \text{if no sale} \\ 500 & \text{if sale B} \\ 1000 & \text{if sale A} \end{cases}$$

$\therefore$ X can take values

$\{0, 500, 1000, 1500, 2000\}$

[Different ways of adding values of S_1 and S_2]

from Question we know that

$P(S_1 = 0) = 1 - P(\text{sale in } S_1)$

$= 1 - 0.3 = 0.7$

$P(S_1 = 500) = \frac{1}{2} \times P(\text{Sale in } S_1)$ $\qquad$ Since A and B are equally likely

$= 0.15$

$P(S_2 = 1000) = \frac{1}{2} \times P(\text{Sale in } S_1)$

$= 0.15$

Similarly

$P(S_2 = 0) = 0.4$

$P(S_2 = 500) = 0.3$

$P(S_2 = 1000) = 0.3$

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S_2 \ S_1	0	500	1000
0	0	500	1000
500	500	1000	1500
1000	1000	1500	2000

This table shows which combination of S_1 and S_2 gives us each value of random variable X

Since S_1 and S_2 are given to be independent,

$$P(S_1 = 500 \cap S_2 = 500) = P(S_1 = 500) \cdot P(S_2 = 500) = 0.045$$

etc for all combinations

S_2 \ S_1	0	500	1000
0	0.28	0.06	0.06
500	0.21	0.045	0.045
1000	0.21	0.045	0.045

∴ PMF of X is given by adding the probabilities of corresponding values of x

$$P_X(x) = \begin{cases} 0.28 & x = 0 \\ 0.21 + 0.06 & x = 500 \\ 0.21 + 0.045 + 0.06 & x = 1000 \\ 0.045 + 0.045 & x = 1500 \\ 0.045 & x = 2000 \end{cases}$$

$$P_X(x) = \begin{cases} 0.28 & x = 0 \\ 0.27 & x = 500 \\ 0.315 & x = 1000 \\ 0.09 & x = 1500 \\ 0.045 & x = 2000 \end{cases}$$

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